

Book-A-Doctor Healthcare Appointment Management System

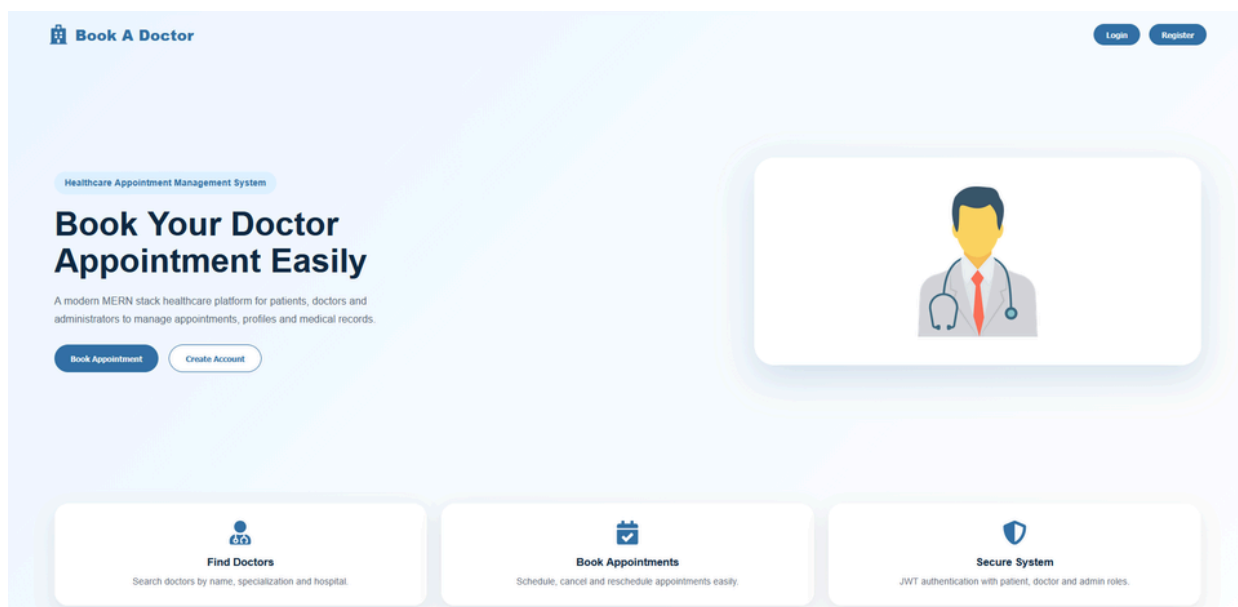
1. Introduction

ProjectTitle: **Book-A-Doctor**– Healthcare Appointment Management System.

This project is a full-stack web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The system is designed to bridge the gap between patients and healthcare providers by providing a centralized online platform for appointment management.

The application supports three major user roles: Patient, Doctor, and Administrator. Patients can search for doctors, book appointments, manage their profiles, upload medical records, and track appointment history. Doctors can manage appointments, review patient information, update appointment status, and maintain professional profiles. Administrators oversee the platform by approving doctors, monitoring system activity, and managing users.

Cover Page:



2. Project Overview

Purpose:

The primary purpose of the Book-A-Doctor system is to simplify the healthcare appointment process. Traditional appointment systems often involve phone calls, manual registers, and long waiting periods. This project digitizes the entire workflow.

Goals:

- Reduce appointment scheduling time.
- Improve accessibility to healthcare services.
- Provide secure digital management of appointments.
- Enable doctors to manage schedules efficiently.
- Provide administrative control over the platform.

Major Features:

- Patient Registration and Login
- Doctor Registration and Login
- Admin Approval Workflow
- Doctor Search and Filtering
- Appointment Booking
- Appointment Cancellation and Rescheduling
- Medical Records Upload
- Profile Management
- Dashboard Analytics
- Secure JWT Authentication

Patient Dashboard:

The screenshot displays the Patient Dashboard interface. On the left is a dark blue sidebar titled "Patient Panel" with menu items: Profile, Doctors, Book Appointment, My Appointments, Reports, Notifications, Change Password, and Logout. The main content area has a header "Patient Dashboard" with a subtitle "Manage appointments, medical reports and doctor bookings". Below the header are four summary cards: "Total Doctors" (1), "Appointments" (3), "Reports" (0), and "Notifications" (2). The dashboard is divided into two main sections. The left section, "Profile Management", contains input fields for "Name" (filled with "vinay"), "Email" (filled with "vinay@gmail.com"), and "Phone Number" (filled with "987654321"), along with a "Medical History" text area and an "Update Profile" button. The right section, "Profile Picture", shows a message "No profile picture added" and a "Post Image URL" input field, with an "Update Picture" button. At the bottom is a "Doctor Search" section with input fields for "Search by doctor name", "Specialization", and "Hospital", and a "Search" button. Below the search fields, a search result for "Dr. Krishna" is displayed, showing "Specialization: primary care", "Hospital: Not added", and "Consultation Fees: ₹0".

Doctor Panel

Profile

Availability

Appointments

Patients

Prescriptions

Reports

Change Password

Logout

Welcome Dr. krishna

Manage appointments, patients, availability and prescriptions

Total Patients

3

Total Appointments

3

Completed

0

Pending

1

Profile Management

Phone

primary care

Hospital

Qualifications comma separated

Experience

Consultation Fees

Save Profile

Availability Management

Available Days e.g. Monday, Tuesday

Time Slots e.g. 10:00-12:00, 14:00-16:00

Leave Days e.g. 2026-06-20, 2026-06-25

Save Availability

Appointment Management

Patient	Date	Time Slot	Status	Reason	Actions
vinay	15/6/2026		approved	fever	CompleteReschedule
vinay	15/6/2026	10:00	approved	cold	CompleteReschedule

Patient	Date	Time Slot	Status	Reason	Actions
vinay	15/6/2026		Completed	fever	
vinay	15/6/2026	10:00	Approved	cold	Complete Reschedule
shruthi	15/6/2026	10:00	Pending	fever	Accept Reject

Name: vinay Last Reason: fever Status: completed Medical history and uploaded reports can be expanded after adding full reports backend.	
Name: vinay Last Reason: cold Status: approved Medical history and uploaded reports can be expanded after adding full reports backend.	
Name: shruthi Last Reason: fever Status: pending Medical history and uploaded reports can be expanded after adding full reports backend.	

Patient Name	
Medicine	
Dosage	
Instructions	
Add Prescription	Download Prescription

Total Patients: 3
Total Appointments: 3
Monthly Appointments: 3
System Doctors: 3

[Change Password](#)

Admin Panel

Statistics

Approvals

Doctors

Patients

Appointments

Specializations

Settings

Logout

Add Doctor

Patient Management

Name	Email	Phone	Medical History	Actions
John Doe	patient@example.com	Not added	None	Delete
vinay	vinay@gmail.com	987654321	None	Delete
shruthi	shruthi@gmail.com	Not added	None	Delete
v	v@gmail.com	Not added	None	Delete
k	k@gmail.com	Not added	None	Delete
vinayc	vinayc@gmail.com	Not added	None	Delete
v1	v1@gmail.com	123456789	None	Delete
k1	k1@gmail.com	Not added	None	Delete
Latha	Latha@gmail.com	Not added	None	Delete

Appointment Management

Admin Panel

Statistics

Approvals

Doctors

Patients

Appointments

Specializations

Settings

Logout

Appointment Management

Patient	Date	Time Slot	Status	Reason	Actions
	15/6/2026		approved		Approve Complete Cancel
John Doe	12/6/2026		approved	Checkup	Approve Complete Cancel
John Doe	15/6/2026		pending	General Checkup	Approve Complete Cancel
vinay	15/6/2026		pending	fever	Approve Complete Cancel
vinay	15/6/2026		completed	fever	Approve Complete Cancel
vinay	15/6/2026	10:00	approved	cold	Approve Complete Cancel
shruthi	15/6/2026	10:00	pending	fever	Approve Complete Cancel

Specialization Management

Add Specialization

Add Specialization

Cardiologist

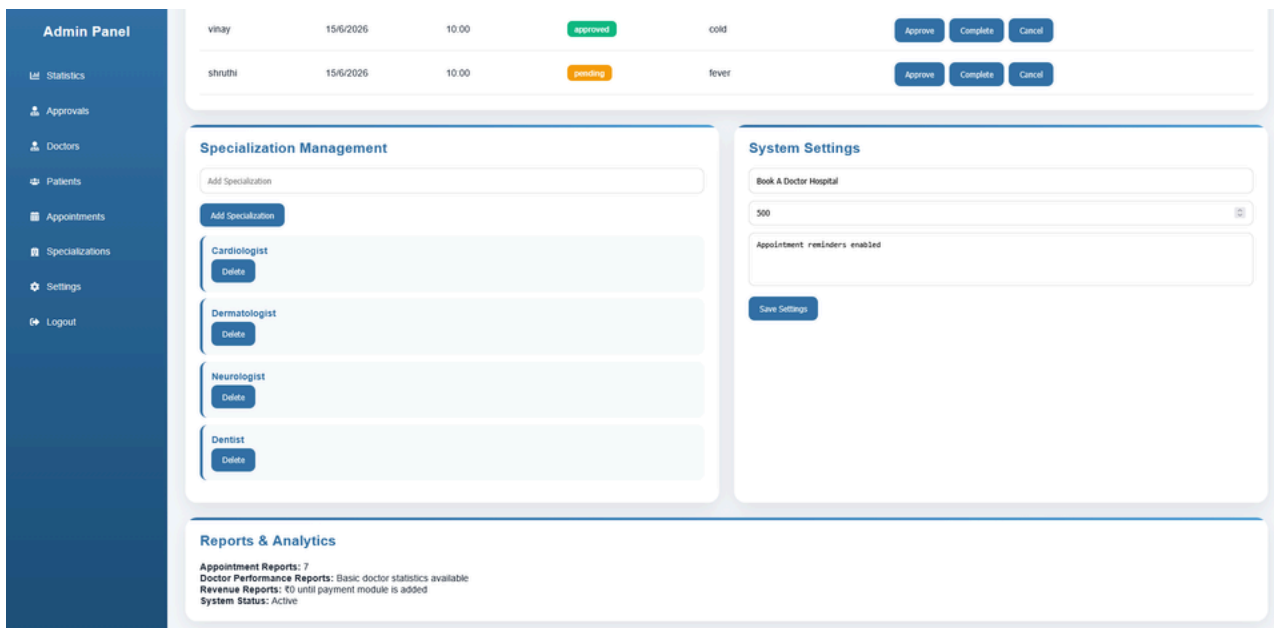
Delete

System Settings

Book A Doctor Hospital

500

Appointment reminders enabled



3. Architecture

Frontend Architecture:

The frontend is developed using React.js. React Router is used for navigation. Axios is used for API communication. Components are separated into reusable modules to improve maintainability.

Backend Architecture:

The backend is built using Node.js and Express.js. RESTful APIs are used to handle communication between frontend and database layers. Middleware is used for authentication and authorization.

Database Architecture:

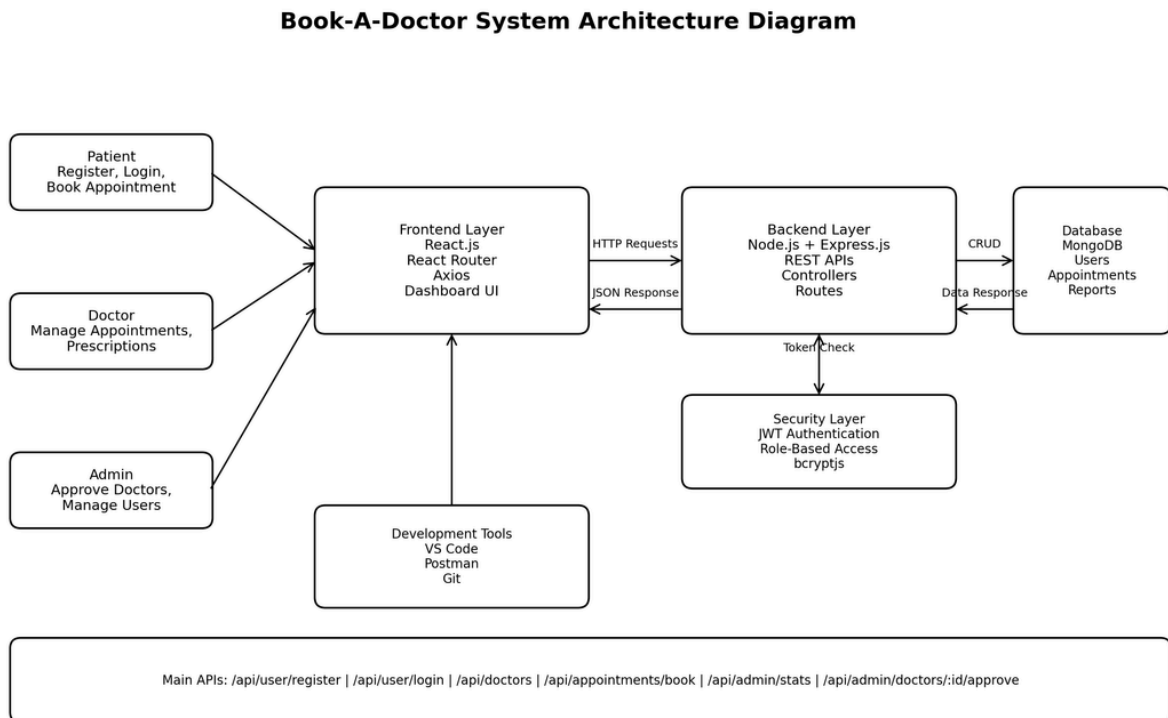
MongoDB serves as the primary database. Collections include:

- Users
- Appointments
- Medical Records
- Notifications

System Flow:

User → React Frontend → Express APIs → MongoDB Database → Response → Frontend

System Architecture Diagram:



4. Setup Instructions

Prerequisites:

- Node.js
- MongoDB
- Git
- VS Code
- Postman

Installation Steps:

Step 1:

Clone the repository.

Step 2:

Navigate to the server folder and install dependencies.

`npm install`

Step 3:

Navigate to the client folder and install dependencies.

`npm install`

Step 4:

Configure environment variables:

`PORT=5000 MONGO_URI=mongodb://127.0.0.1:27017/doctorApp JWT_SECRET=secretkey`

Step 5:

Start Backend:

`npm start`

Step 6:

Start Frontend:

`npm start`

The application becomes accessible through localhost URLs.

5. Folder Structure

Client Folder:

- src
- components
- pages
- styles
- api
- assets

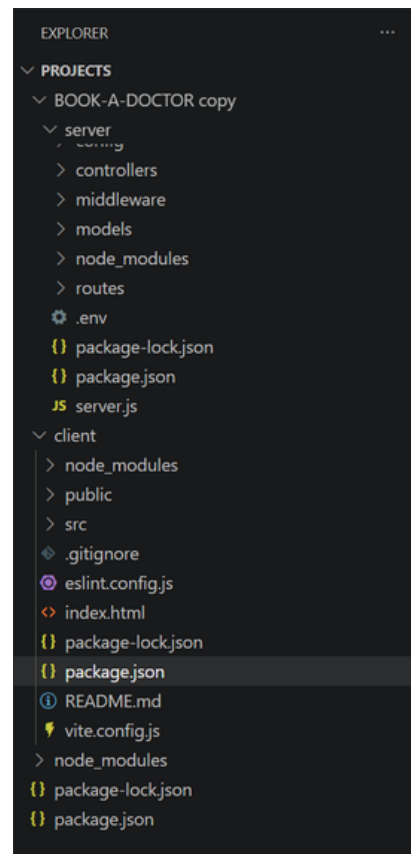
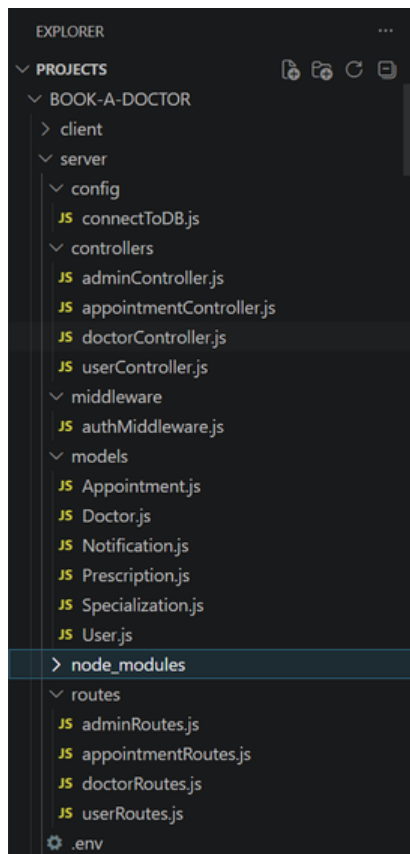
Server Folder:

- controllers
- models
- routes
- middleware
- config
- utils

Key Files:

- server.js
- package.json
- authMiddleware.js
- appointmentController.js
- userController.js

Project Folder Structure



6. Running the Application

Backend:

```
cd server  
npm start
```

Frontend:

```
cd client  
npm start
```

Workflow:

1. Start MongoDB.
2. Run backend server.
3. Run frontend application.
4. Open browser.
5. Login as Patient, Doctor, or Admin.

Expected Outcome:

The system loads dashboard pages according to the authenticated user role.

7. API Documentation

User APIs:

```
POST /api/user/register  
POST /api/user/login  
GET /api/user/profile  
PUT /api/user/profile  
PUT /api/user/change-  
password
```

Doctor APIs:

```
GET /api/doctors  
GET  
/api/doctors/appointments
```

Appointment APIs:

```
POST /api/appointments/book  
GET /api/appointments/patient/:id  
GET /api/appointments/doctor/:id  
PUT /api/appointments/:id/status  
PUT /api/appointments/:id/reschedule
```

Admin APIs:

GET /api/admin/doctors

GET /api/admin/patients

GET /api/admin/stats

PUT /api/admin/doctors/:id/approve

PUT /api/admin/doctors/:id/reject

The APIs are protected using JWT authentication and role-based authorization.

8. Authentication

Authentication Mechanism:

The application uses JSON Web Tokens (JWT) for secure authentication.

Process:

1. User logs in.
2. Credentials are validated.
3. JWT token is generated.
4. Token is stored in local storage.
5. Protected APIs verify the token.

Authorization:

- Patient Access
- Doctor Access
- Admin Access

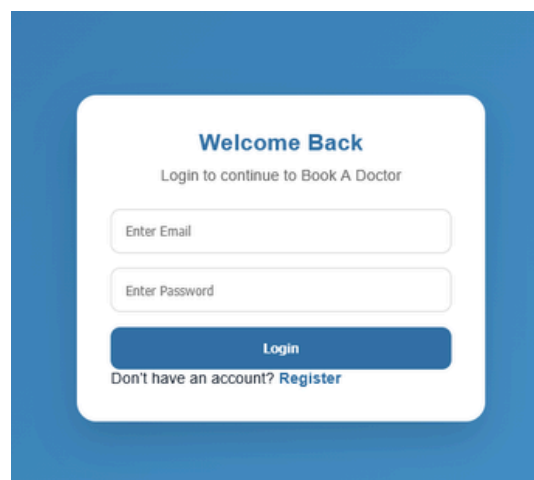
Password Security:

Passwords are encrypted using bcryptjs before being stored in MongoDB.

Benefits:

- Secure communication
- Role-based access control
- Prevention of unauthorized access

Login Page:

The image shows a login page with a blue background. In the center is a white rounded rectangle containing the login form. At the top of the form, it says "Welcome Back" in blue, followed by "Login to continue to Book A Doctor" in a smaller font. Below this are two input fields: "Enter Email" and "Enter Password". Under the password field is a blue "Login" button. At the bottom of the form, it says "Don't have an account? Register" with "Register" as a link.

9. User Interface

Patient Interface:

- Dashboard
- Doctor Search
- Appointment Booking • Appointment History
- Profile Management
- Medical Records

Doctor Interface:

- Dashboard
- Appointment Management
- Patient Records
- Profile Settings

Admin Interface:

- Dashboard
- Doctor Approval
- User Management
- Statistics

Recommended Screenshots:

1. Login Page
2. Registration Page
3. Patient Dashboard
4. Doctor Dashboard
5. Admin Dashboard
6. Book Appointment Page
7. Search Doctor Page
8. Medical Records Page
9. Profile Update Page

10. Testing

Testing Strategy:

Functional Testing:

- Registration
- Login
- Doctor Search
- Appointment Booking
- Appointment Rescheduling

Integration Testing:

- Frontend-Backend communication
- Database integration

Security Testing:

- JWT Validation
- Protected Routes
- Role Validation

Sample Test Cases:

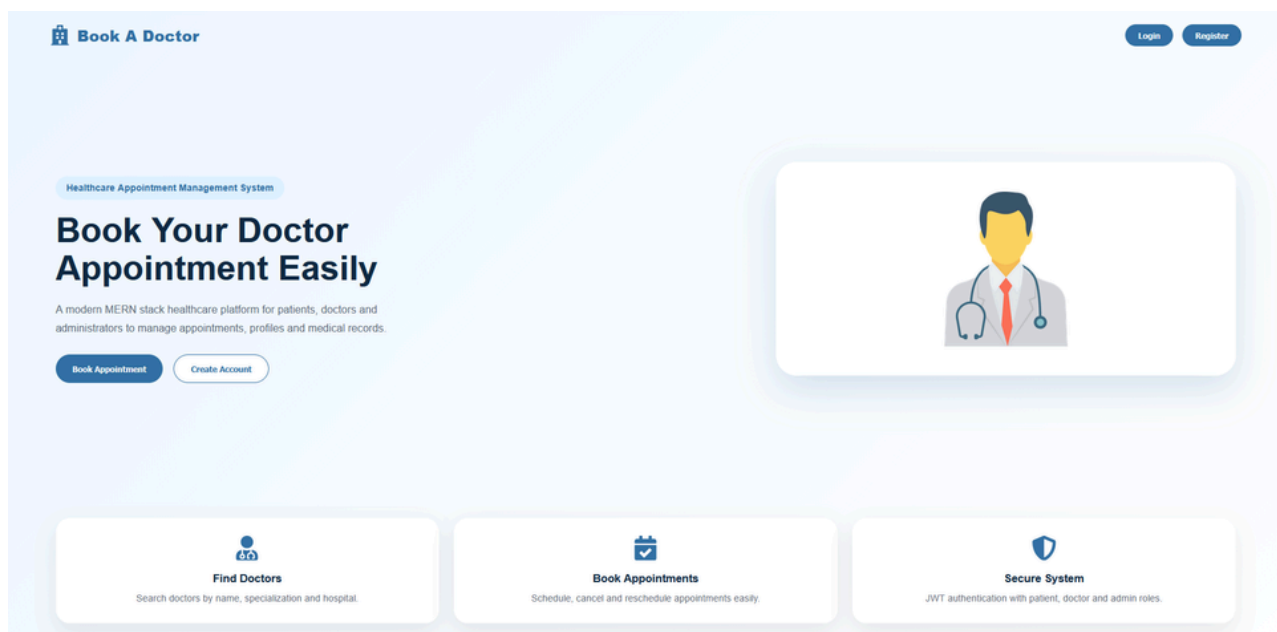
1. Valid Login
2. Invalid Login
3. Appointment Booking
4. Doctor Approval
5. Profile Update

Expected Result:

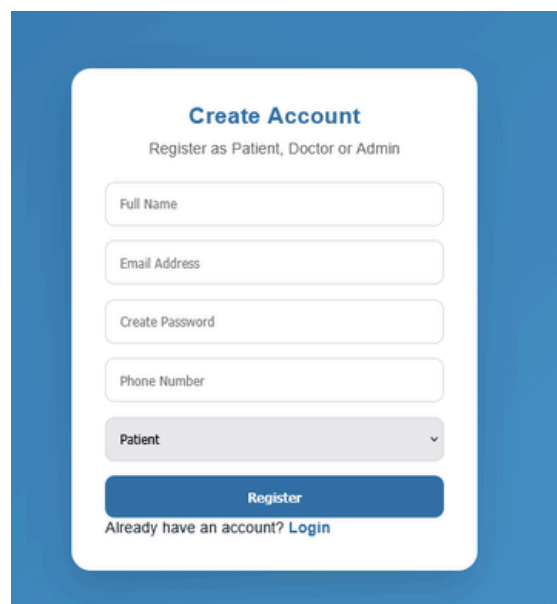
All major modules perform correctly under normal conditions

11. Screenshots or Demo

1. Home Page:



2. User Registration:

The screenshot shows a "Create Account" registration form. The form is titled "Create Account" and has a subtitle "Register as Patient, Doctor or Admin". It contains five input fields: "Full Name", "Email Address", "Create Password", "Phone Number", and a dropdown menu for "Patient". Below the input fields is a blue "Register" button. At the bottom, there is a link "Already have an account? Login".

3. User Login:

Welcome Back

Login to continue to Book A Doctor

Login

Don't have an account? [Register](#)

4. Patient Dashboard:

Profile

Dashboard

Book Appointment

My Appointments

Reports

Notifications

Change Password

Login

Patient Dashboard

Manage appointments, medical reports and doctor bookings

Total Doctors

1

Appointments

3

Reports

0

Notifications

2

Profile Management

Full Name

Email

Phone Number

Medical History

Save Profile

Profile Picture

No profile picture added

Doctor Search

Dr. Amina

Specialization: General

Specialty: General

Specialty: General

Profile

Dashboard

Book Appointment

My Appointments

Reports

Notifications

Change Password

Login

Book Appointment

Book a doctor appointment

My Appointments

Date	Time Slot	Status	Reason	Actions
15/05/2025		Pending	Test	Cancel Reschedule
15/05/2025		Completed	Test	Cancel Reschedule
15/05/2025	10:00	Completed	Test	Cancel Reschedule

Medical Reports

No reports uploaded

Notifications

Appointment Updates
Appointment confirmations and cancellations will appear here

Reminder
Please check your booked appointments regularly

Profile

Dashboard

Book Appointment

My Appointments

Reports

Notifications

Change Password

Login

My Appointments

Manage appointments

Date	Time Slot	Status	Reason	Actions
15/05/2025		Pending	Test	Cancel Reschedule
15/05/2025		Completed	Test	Cancel Reschedule
15/05/2025	10:00	Completed	Test	Cancel Reschedule

Medical Reports

No reports uploaded

Change Password

Notifications

Appointment Updates
Appointment confirmations and cancellations will appear here

Reminder
Please check your booked appointments regularly

5. Doctor Dashboard:

Doctor Panel

Profile

Availability

Appointments

Patients

Prescriptions

My Reports

Change Password

Logout



Welcome Dr. Krishna

Manage appointments, patients, availability and prescriptions

Total Patients

3

Total Appointments

3

Completed

0

Pending

1

Profile Management

Phone

email ID

password

confirm password (retype)

Save Profile

Availability Management

Available From e.g. Monday, Tuesday

From Time e.g. 10:00 (12:00, 18:00, 20:00)

Leave From e.g. 20:00 (06:00, 20:00-22)

Save Availability

Appointment Management

Patient	Date	Time Slot	Status	Reason	Actions
AKSH	15/03/2025		Completed	None	Details Reschedule
AKSH	15/03/2025	10:00	Completed	None	Details Reschedule

Doctor Panel

- Profile
- Availability
- Appointments
- Patients
- Prescriptions
- Reports
- Change Password
- Logout

Patient Name	Date	New Visit	History	Message	Action
John Doe	1/10/2025	<button>Update</button>			
Jane Smith	1/10/2025	<button>Update</button>	<button>View</button>	<button>Send</button>	<button>Cancel</button> <button>Reschedule</button>
Alice Johnson	1/10/2025	<button>Update</button>	<button>View</button>	<button>Send</button>	<button>Add</button> <button>Delete</button>

Patient Management

Name: John Doe | Last Revision: None | History: Confirmed

Medical history and updated reports can be expanded after setting full reports backend.

Name: Jane Smith | Last Revision: None | History: Reported

Medical history and updated reports can be expanded after setting full reports backend.

Name: Alice Johnson | Last Revision: None | History: Pending

Medical history and updated reports can be expanded after setting full reports backend.

Prescription Management

Patient Name:

Disease:

Medicine:

Instructions:

+ Add New Prescription
+ Update Existing Prescription

Reports & Analytics

Total Patients: 1

Total Appointments: 1

Monthly Appointments: 1

Active Patients: 1

[Change Password](#)

The image shows a web application interface for a doctor's panel. The sidebar on the left contains navigation links: Profile, Availability, Appointments, Patients, Prescriptions, Reports, Change Password, and Logout. The main content area is divided into two primary sections: Patient Management and Prescription Management.

Patient Management displays a table with the following data:

Name	Age	Last Reason	Score
John Doe	35	Headache	85
Jane Smith	28	Stomach Issues	72
Robert Johnson	42	Joint Pain	68

Below the table, a note states: "Medical history and updated reports can be expanded after adding full reports backend."

Prescription Management includes a form with the following fields:

- Prescription Name:** A text input field.
- Medication:** A text input field.
- Dosage:** A text input field.
- Frequency:** A text input field.

At the bottom of this section are two buttons: "Add Prescriptions" and "Updated Prescriptions".

Reports & Analytics section shows a summary of statistics:

- Total Patients: 12
- Total Appointments: 25
- Monthly Appointments: 18
- System Downtime: 0

The **Change Password** section at the bottom contains three input fields for "Old Password", "New Password", and "Confirm Password", followed by a "Save Password" button.

6. Admin Dashboard:

[illegible]

Patient Management					
Name	Email	Phone	Medical History	Actions	
John Doe	john.d@example.com	Not added	None	View	
Jane Doe	jane.d@gmail.com	987654321	None	View	
Smith	smith@gmail.com	Not added	None	View	
E	e@gmail.com	Not added	None	View	
	e@gmail.com	Not added	None	View	
Smith	smith@gmail.com	Not added	None	View	
E	e@gmail.com	123456789	None	View	
	e@gmail.com	Not added	None	View	
Latha	Latha@gmail.com	Not added	None	View	

[illegible][illegible]

7. Doctor Search:

Doctor Search

Search

Name: Dr. krishna
Specialization: primary care
Hospital: Not added
Consultation Fees: ₹0

8. Appointment Booking:

Book Appointment

Dr. krishna - primary care

16 / 06 / 2026

10:30

Fever and cold

Book Appointment

9. Appointment List:

My Appointments				
Date	Time Slot	Status	Reason	Actions
15/6/2026		pending	fever	Cancel Reschedule
15/6/2026		completed	fever	
15/6/2026	10:00	approved	cold	Cancel Reschedule

10. Profile Management:

Profile Management

vinay

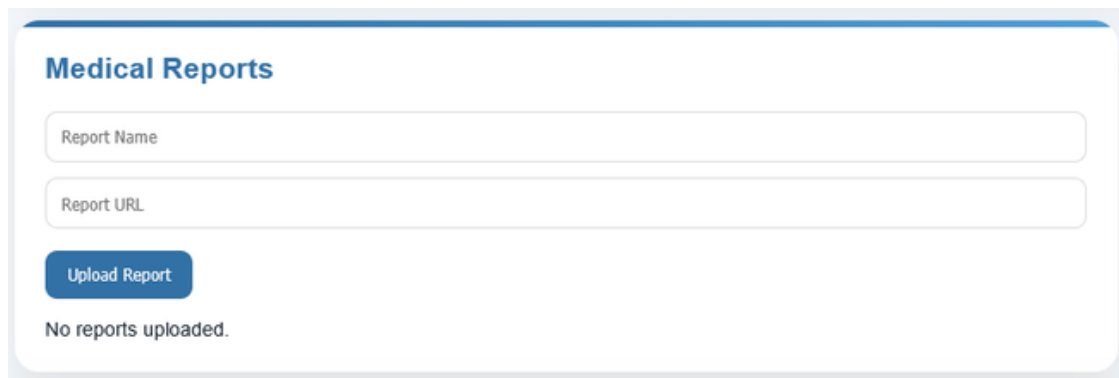
vinay@gmail.com

987654321

NA

Update Profile

11. Medical Records Upload:

A screenshot of a web form titled "Medical Reports". It features two input fields: "Report Name" and "Report URL". Below these fields is a blue button labeled "Upload Report". At the bottom of the form, it says "No reports uploaded.".

Medical Reports

Report Name

Report URL

Upload Report

No reports uploaded.

12. Known Issues

Current Limitations:

1. No Payment Gateway:

Payments are not integrated.

2. Email Notifications:

Email and SMS notifications are not implemented.

3. Real-Time Availability:

Doctor availability updates may require refreshing.

4. Scalability:

Additional optimization may be required for large-scale deployments.

13. Future Enhancements

Future Scope:

1. Online Video Consultation.
2. AI-Based Doctor Recommendation System.
3. Payment Gateway Integration.
4. Email and SMS Notifications.
5. Mobile Application Development.
6. Electronic Health Records Integration.
7. Multi-Hospital Management.
8. Real-Time Appointment Slot Locking.
9. Advanced Analytics Dashboard.

10. Doctor Ratings and Reviews.
11. Chat Support Between Patient and Doctor.
12. Cloud Deployment with CI/CD.

Conclusion:

The Book-A-Doctor project successfully demonstrates a complete healthcare appointment management solution using modern web technologies. The system provides secure authentication, appointment scheduling, doctor management, and administrative control while offering a scalable foundation for future healthcare innovations.